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REGULATORY AND COMPLIANCE FILE – EEG FIELD KIT

Document: REG-EEG-012 **Revision:** B **Date:** 2026-09-01 **Issued by:** TI One Voice research programme (one.witysk.org), Brussels, Belgium **Licence:** CC BY-SA 4.0 **Governing documents:** DSN-EEG-003 Rev C, then RFQ-EEG-001 Rev E. Where this document and design.py disagree, design.py governs.

Rev B in one line: the carrier is now 150.0 x 130.0 mm on **four** layers and POD-P1 has grown with it, so the RoHS scope, the EUT description and the creepage clause are restated; every production-test citation is taken from TST-EEG-004 Rev C rather than invented; the HM-07 / HM-08 / HM-10 part names are corrected; there are **fourteen** patient terminations, not thirteen; the lithium section keeps the obligation and hands the procedure to PKG-EEG-015 section 7; and S-02, E-29, F-06, A-04 and the serial-number format are brought into line with the package rulings of 2026-09-01. **The findings of the second cross-document audit of 1 September 2026 are closed in this issue**: clause 8.5.2's creepage argument now cites the DRC report as its evidence instead of asserting the premise, S-04 and E-23 are separated and each stated at its true status, and every document cited here carries its current revision letter. **Corrections within Rev B, after the verification review of package v2.1 of 1 September 2026**: section 3.1 now says which documents actually match the no-spare-cell decision and which one does not, section 3.4 carries the CASE-00 Rev C seven-layer foam insert in place of the withdrawn two-sheet one, section 3.8 hands the whole cell-replacement policy to SVC-EEG-013 section 2 R9 instead of stating a second and different one, and section 6.1's reuse figure is restated from SVC-EEG-013 section 3.5 and RISK-EEG-011 section 1.4. **Corrections within Rev B, after the routing closed on 2026-09-01**: the DRC report now records zero violations with all 145 nets connected and both inner planes continuous, so the introduction, section 4.2, clause 8.5.2, section 8 item 18 and section 9 items 1 and 10 are restated – the fabrication data is released for review under RFQ-EEG-002A and is still not released for fabrication, because no human layout engineer has reviewed the routing, which closed 169 connections at relaxed geometry. S-02, S-04 and E-23 are untouched by the routing and stay exactly as section 9 records them. **Corrections within Rev B, after the independent review of package v2.2 of 2026-09-02**: section 3.4 is restated against what `mech/` now holds – the seven CASE-00 Rev C layer cut files are shipped and the Rev B pair is deleted – while keeping the true half of that row, which is that they are not released for cutting and no insert has been cut; and section 9's simulation figures are quoted from the run of `tools/simulate_production.py` of 2 September 2026 rather than from an earlier one. **Corrections within Rev B, after the design and firmware changes of 2026-09-02**: three statuses this file carried as not met are now met in the design and are corrected in place, with their date, wherever they appear – **S-02** (ECO-EEG-024 applied, R1-R16 are 68 kΩ, single-fault DC 36.8 µA against 50 µA: clause 4.7, clause 8.4.2, section 8 item 21 and section 9 item 5), **E-27** (the bicolour contact-light phase driver is written: section 4.3 mode M5, section 8 item 18, clause 14) and **E-11** (the Sallen-Key moved from X7R to C0G). The firmware has also been **built** for the first time, against ESP-IDF v5.2.5, and has booted once under QEMU; clause 14 is restated for that. **None of these is a measurement, a test or an approval**: no unit exists, nothing has run on hardware, no fault-insertion or emissions testing has been done, and no safety or notified reviewer has looked at any of it. The revision letter of this file does not change; these are corrections within the same release.

Why this document exists

The v1 package carried its entire regulatory position in eleven words on a label and two one-line requirements. “Research instrument – not a medical device” appeared twice, in RFQ-EEG-001 Rev C M-03 and in DSN-EEG-002 section 7, with no declaration, no reasoning and no controlled wording; S-05 asked for ISO 10993 declarations without naming a material, a supplier or an acceptance criterion; S-06 asked for a CISPR 11 pre-scan with no test plan and no place in production; S-08 asked for a RoHS and REACH declaration with no template; and S-04 asked for a UN 38.3 report to be “available” for the cell. Nothing anywhere in the v1 package mentioned how a box containing a lithium cell gets from a factory to Brussels, to a stranger’s home and back again. The 14-agent audit of v1 raised that as a blocking finding (`lithium-shipping-dossier`) alongside `not-a-medical-device-declaration`, `rohs-reach-declarations`, `emc-prescan-plan`, `biocompatibility-dossier`, `cell-qualification-evidence` and `ethics-dpia-configuration-link`. This file closes all seven. It states what the programme is legally obliged to do, who does it, what evidence is kept, and what has not been done.

Two things changed in the hardware while package v2 was being laid out, and both reach this file. The carrier grew from 130 x 124 mm to **150.0 x 130.0 mm**, because thirty connectors, 211 parts and 156 nets would not close at the smaller size. And the carrier went from two layers to **four** – L1 signal, L2 reference plane, L3 reference plane, L4 signal. Package v1 asserted that a two-layer carrier would be cheap and easy to route; actually doing the layout showed that it is not, because on two layers the bottom side has to be both the reference plane and the second routing surface and it cannot be both. Four layers give two full routing surfaces and a continuous reference under every analogue trace, which is what the isolation, star-point and reference-plane rules of DSN-EEG-003 section 3.3 require and what a swiss-cheesed two-layer pour cannot deliver. The enclosure grew to match: POD-P1 base 163.0 x 143.0 x 58.0 mm external, 158.0 x 138.0 x 55.5 mm internal, and the MP-01 module plate 146.0 x 126.0 x 3.0 mm. Three things in this file follow from that and are marked where they occur: the RoHS declaration now has to cover a multilayer laminate rather than a double-sided one (section 2.2 and 2.3), the equipment under test for the EMC pre-scan is a different physical object (section 4.2), and the isolation keep-out that carries the creepage argument has to be copper-free on **four** layers rather than two (section 5, clause 8.5.2). That last one is a fact about the artwork, so it is checked against the DRC report and quoted from it rather than asserted.

The routing exists, it closes, and it is released for review and not for fabrication. `kicad/EEG-CAR-01_RevB_DRC_report.txt` is the authority for the state of the layout, and this file quotes it rather than paraphrasing it. It records a four-layer 150.0 x 130.0 mm board with 3 745 track segments and 552 through vias, each reference plane one continuous island per net, **all 145 nets fully connected** and **zero violations of any rule**. That meets all three conditions of the fabrication-release gate in ECO-EEG-016 section 3, so the data in `kicad/` is **released for review under RFQ-EEG-002A**. It is **not released for fabrication**: the routing was produced by the programme’s own tools, no human layout engineer has looked at it, and the router closed 169 connections at relaxed geometry. Section 9 item 10 carries the full figures. Nothing in this file may be read as saying that a board has been made or that anything on it has been measured.

Nothing in this package has been manufactured, shipped, scanned or tested. **No safety engineer has reviewed this design.** Every limit below is an acceptance criterion, not a result. Where a figure is derived rather than measured it is labelled *calculated*.

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1. Regulatory status

1.1 The declaration

The TI One Voice EEG field kit is a research instrument built by, and for, a single pre-registered study. It is **not a medical device within the meaning of Regulation (EU) 2017/745 (MDR)**, it is **not placed on the market**, and it bears **no CE mark**.

The instrument is loaned, never sold and never transferred in ownership. It is issued only to a person who has consented to a specific study protocol, is used only for the recording sessions that protocol defines, and is returned. It returns no result of any kind to the participant during or after a session. The study's findings are published as pre-registered group results.

1.2 The reasoning, written out

MDR Article 2(1) defines a medical device by its intended purpose as stated by the manufacturer. The programme is the manufacturer, and the intended purpose it states is: *to acquire sixteen channels of electrophysiological signal with co-registered audio for the purposes of a research study, under a protocol approved by an independent ethics committee*. That purpose contains no diagnosis, no prevention, no prediction, no prognosis, no monitoring of a disease or condition in an individual, no treatment and no alleviation. Nothing the device produces is looked at by a clinician, is entered in anyone's health record, or is returned to the individual as information about themselves.

Two further points have to be handled explicitly, because a reviewer will ask about both.

MDR Article 5(5), in-house manufacture. This is the exemption a health institution uses when it makes and uses a device on its own patients. It does not apply here and is not relied upon: the programme is not a health institution, and the participants are not its patients. The programme's position is that the device falls outside MDR entirely on the intended-purpose test of Article 2(1), so no exemption is needed. Article 5(5) is named here only so that the record shows it was considered and set aside.

Clinical investigation. MDR Articles 62 and 82 govern clinical investigations of devices. Article 62 covers investigations run to demonstrate conformity for a CE mark; that is not what this is, because no CE mark is sought and no conformity claim is made. Article 82 covers other clinical investigations of devices, and its detailed rules are set by national law. The study is a research study using an instrument, not an investigation of the instrument. **This determination is the ethics committee's to make, not the programme's**, and it is submitted to the committee as an explicit question in ETH-EEG-001. If the committee or the Belgian competent authority classifies the study otherwise, this section is revised by ECO before any unit is issued to a participant.

1.3 Reconciliation with IEC 60601-1

RFQ-EEG-001 Rev E section 8 says the instrument is designed to the applicable principles of IEC 60601-1 for a type BF applied part, and Annex A lists IEC 60601-1 and IEC 60601-2-26 as design references. Read carelessly, that looks like a medical claim sitting next to a denial of one. It is not. The device is connected to a person's scalp, so the programme borrows the engineering discipline that exists for exactly that situation:

series resistance in every electrode lead, a galvanic barrier to any mains-referenced host, battery-only recording, touch-proof electrode connectors. Borrowing the design principles of a standard is not a claim of conformity to it, and conformity to IEC 60601-1 would not by itself make anything a medical device. Section 5 below sets out, clause by clause, which of those principles the design actually meets and which it does not.

1.4 Controlled wording

These strings are controlled. They change only by ECO under ECO-EEG-016, and any change re-triggers the label artwork in PKG-EEG-015 and a notification to the ethics committee.

Where	Exact text	Owner
Enclosure label, line 4 (ART-LBL-01, per ASM-EEG-007 stage 6 and RFQ M-03)	`RESEARCH INSTRUMENT -- NOT A MEDICAL DEVICE`	PKG-EEG-015 artwork
Participant quick-start card (RFQ A-06)	“This is a research instrument, not a medical device. It cannot detect, confirm or rule out anything about your health or your experiences, and nobody will read your recording as a test result.”	IFU-EEG-014
Participant information sheet and consent form	“The headset records electrical activity from your scalp and sound from a microphone. It is a research instrument. It is not a medical device, it is not a diagnostic test, and the recording will not tell you or anyone else whether what you experience is real, imagined, or caused by anything in particular. You will not be given an individual result.”	ETH-EEG-001, ethics submission
Commercial invoice and customs declaration	“Non-medical research instrument, loaned scientific equipment, not for sale, not for clinical use. Value declared for customs purposes only.”	Section 1.5
Website, repository README, public posts	“A research EEG instrument built for one pre-registered study. It is not a medical device and gives no individual result.”	Programme lead

Where	Exact text	Owner
Manufacturer's own material	The bidder may not describe this instrument as medical, clinical, diagnostic or medical-grade in any catalogue, case study, website or tender reference. This clause goes in the purchase order terms	Purchase order

Prohibited words anywhere in participant-facing or public material: *diagnostic, diagnosis, clinical, medical-grade, therapy, therapeutic, treatment, screening, patient, monitor* in a clinical sense. Use *participant*, not *patient*. The single retained exception is the standard's own term "patient auxiliary current" in RFQ S-02 and in section 5 below, where changing it would misname a defined quantity.

1.5 Customs

Tariff classification and regulatory status are different questions and must not be conflated. An EEG amplifier classifies under HS heading 9018 (instruments and appliances used in medical, surgical, dental or veterinary sciences), most plausibly 9018.19, and that is a duty and statistics question decided by the declarant and the customs broker. Declaring 9018 does not make the instrument a medical device under MDR, and the invoice text in section 1.4 says so on the same page.

> **Open.** The HS code has not been confirmed with a broker. Getting it wrong risks a health-authority hold on arrival in Brussels, which is exactly the delay a Phase 2 build cannot absorb. The programme confirms the code with its broker before the first international despatch. Owner: programme lead.

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2. RoHS 3 and REACH

2.1 What actually applies

RoHS (Directive 2011/65/EU as amended by (EU) 2015/863, commonly "RoHS 3") is triggered by placing electrical and electronic equipment on the market. The kit is not placed on the market, so RoHS is not a legal obligation on the programme. The research-and-development exclusion in Article 2(4)(j) is *not* relied on either, because it is limited to equipment made available business-to-business and these kits go to private individuals. The programme therefore requires RoHS 3 conformity **contractually**, through RFQ S-08, and holds the evidence in the same form the Directive would require. The reason is practical: partner institutions, university compliance offices and the ethics committee ask for it, and a fleet that cannot answer is a fleet that stops.

REACH (Regulation (EC) 1907/2006) Article 33 places a communication duty on any supplier of an article containing a candidate-list substance of very high concern above 0.1 % w/w. The programme treats loaning a kit to a participant as supply for this purpose, which is the conservative reading, and therefore holds article-level SVHC information and passes the required statement to the participant in IFU-EEG-014.

2.2 The RoHS declaration the manufacturer completes

One declaration per phase, signed by an officer of the manufacturer, covering every item the manufacturer buys, builds or packs.

Substance	Maximum concentration by weight in a homogeneous material
Lead (Pb)	0.1 %
Mercury (Hg)	0.1 %
Cadmium (Cd)	0.01 %
Hexavalent chromium (Cr VI)	0.1 %
Polybrominated biphenyls (PBB)	0.1 %
Polybrominated diphenyl ethers (PBDE)	0.1 %
Bis(2-ethylhexyl) phthalate (DEHP)	0.1 %
Benzyl butyl phthalate (BBP)	0.1 %
Dibutyl phthalate (DBP)	0.1 %
Diisobutyl phthalate (DIBP)	0.1 %

Required fields: manufacturer legal entity and address; the covered items listed by BOM item number and by carrier reference designator range; the standard applied (**EN IEC 63000:2018**, technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances); the evidence tier per item; every exemption claimed, by Annex III or Annex IV number and expiry date; date; signatory name, position and signature.

The bare board is now a four-layer laminate and the declaration must say so. EN IEC 63000 tests each *homogeneous material* separately, and a four-layer EEG-CAR-01 has more of them than the two-layer board of package v1: the two prepreg plies at 0.200 mm, the 1.065 mm core, the 17 µm inner copper on L2 and L3, the 35 µm outer copper on L1 and L4, the solder mask both sides, the legend ink and the ENIG finish. The fabricator declares the laminate system by trade name and grade, states the flame-retardant chemistry (see section 2.3), and confirms that the declaration covers the inner plies and not only the outer surfaces. A declaration written against a double-sided board is not evidence for this one.

Evidence tiers, in descending order of acceptability: (1) supplier declaration plus material declaration data to IEC 62474; (2) supplier declaration alone; (3) analytical test report from an accredited laboratory. The manufacturer states which tier applies to each item. The programme accepts tier 2 for passives and mechanical parts and requires tier 1 or 3 for anything in section 2.3.

The manufacturer must state, item by item, **which of the twelve purchased module types they can evidence and which they cannot**. Twelve types are bought; **thirteen module assemblies go into each unit**, because the ADS1299 breakout is fitted twice, and the declaration is per type. Consumer breakouts bought from a maker-market vendor frequently carry no RoHS statement at all. An honest “cannot evidence” against the microSD breakout at J20 is acceptable and is recorded as a residual; a blanket “all compliant” covering twelve module types with no supporting documents is not, and is rejected.

2.3 Substances of concern actually present in this BOM

Item	Concern	Position
EEG-CAR-01 bare board, four-layer FR-4 (mask / 35 µm L1 / prepreg 0.200 / 17 µm L2 / core 1.065 / 17 µm L3 / prepreg 0.200 / 35 µm L4 / mask, 1.60 mm ± 10 %)	Brominated flame retardant in the laminate and in both prepreg plies	Standard FR-4 uses TBBPA reacted into the epoxy network, which is **not** a PBB or a PBDE and is therefore outside the RoHS list; the fabricator states this explicitly rather than leaving it to be assumed, names the laminate grade, and declares the mask, the legend ink and the ENIG chemistry. A halogen-free build is not required and is not requested
J15–J17 DIN 42802 sockets, touch-proof 1.5 mm, Stäubli SLB1,5-F class	Lead in the brass or copper-alloy socket body. Lead is on the REACH candidate list (CAS 7439-92-1, added 27 June 2018) and free-machining brass commonly exceeds 0.1 % w/w in that homogeneous material	RoHS exemption 6(c) (copper alloy containing up to 4 % lead) is expected to be claimed. Must be **declared, not assumed**. **The part itself is open:** `design.py` names Stäubli SLB1,5-F as a class, not a confirmed PCB-mount part, and no catalogue part has been confirmed to fit a PCB-mount signal pin with two 1.5 mm retention posts. AVL-EEG-017 carries a 12-week first-article lead time against it, and the declaration cannot be requested until the part is chosen
Harness jacket, headphone cable (item 29), USB cables (item 32), charge pigtail to J24	Phthalates DEHP, DBP, BBP, DIBP in flexible PVC	Require a phthalate-free jacket, or a declaration with test data. This is the single most likely RoHS 3 failure in the kit
Solder in the twelve purchased module types	Lead in solder	Exemptions 7(a) and 7(c)-I may apply to some parts. Modules bought as consumer breakouts are the weak point
HM-01, HM-03, HM-04, **HM-07 boom arm**, **HM-08 battery hatch**, MP-01 module plate and POD-P1 in MJF PA12 dyed black	The dye and any surface treatment	Bureau declares the polymer grade, the colourant and the post-process chemistry. Section 6 needs the same declaration for a different reason, for the printed parts that touch skin

Item	Concern	Position
HM-02 TPU 85A pads and HM-06 chin strap and liner	Plasticiser and colourant in a part held against the forehead and chin for two hours	Tier 1 or 3 evidence required
5 V 2 A EU charger (item 33) and USB cables (item 32), both currently “certified generic”	Nobody knows who made them	See section 2.5. These are the two items in the kit that most need a real CE and RoHS chain and currently have the least
Protected 18650 cell (item 11)	Cadmium and lead limits under the EU Battery Regulation	Supplier RoHS and REACH declarations required with the UN 38.3 summary of section 3
ENIG finish on EEG-CAR-01, Au 0.05–0.10 µm over Ni 3.0–6.0 µm	Nickel release is a REACH Annex XVII restriction for prolonged skin contact	Not applicable: the carrier is inside a closed pod and touches nobody. Recorded so the question is not asked twice

2.4 SCIP

If any article in the kit contains a candidate-list SVHC above 0.1 % w/w, a SCIP notification to ECHA is due from the **EU-established supplier placing the article on the EU market**, which is the programme as importer, not a manufacturer outside the EU. The programme’s position, to be recorded as a decision by the programme lead once the section 2.3 declarations are back:

- If the DIN 42802 socket bodies exceed 0.1 % w/w lead, they are articles in scope and a SCIP notification is prepared, or a lead-free alternate is approved through AVL-EEG-017.
- If no article exceeds the threshold, the decision “no SCIP notification due” is recorded with the declarations that support it, and re-checked at each candidate-list update.

The candidate list is revised roughly twice a year. **Every declaration must state the candidate-list version and date it was checked against.** A declaration that says only “REACH compliant” is worthless and is rejected.

2.5 EU importer position, WEEE, batteries and packaging

At Phase 2 and 3 the programme imports 10 to 50 assembled kits into Belgium from a supplier that may be outside the EU. That makes the programme the importer of the components it does not manufacture, and one of them, the 5 V charger, is a mains-connected consumer good; the USB cables are accessories carried on their own supplier declaration.

Obligation	Position
CE mark on the kit as a whole	**None, and none is applied.** The kit is not placed on the market. See section 1

Obligation	Position
5 V 2 A EU charger (item 33)	The programme requires the actual manufacturer's EU Declaration of Conformity to the Low Voltage Directive 2014/35/EU and the EMC Directive 2014/30/EU, the supporting test reports, and a CE mark traceable to a named legal entity. **"Certified generic" is not acceptable.** If the bidder cannot supply this, the item is replaced with a named-brand charger through AVL-EEG-017
USB cables (item 32)	Named part, supplier declaration, RoHS tier 2 minimum. Two cables ship per kit under RFQ A-07 and **one of them is the host lead**, because the host connection is a socket on the ADuM4160 module and not a captive cable
WEEE, Directive 2012/19/EU	The fleet reaches end of life as electronic waste in Belgium. The programme registers with, or disposes through, the Belgian scheme (Recupel) at end of programme. No WEEE crossed-bin symbol is applied to the kit, because it is not placed on the market; the disposal obligation is discharged directly
Batteries, Regulation (EU) 2023/1542	Take-back and disposal of the fleet's cells through the Belgian collection scheme (Bebat). A cell is never disposed of by a participant. See section 3.8
Packaging, Directive 94/62/EC	Outer carton and PE foam are recyclable, minimised and declared. PKG-EEG-015 owns the carton specification

2.6 Production line audit, from S-08

The bidder answers these on the quotation, as a form rather than as prose:

Question	Answer required
ISO 9001 certificate	Number, certifying body, expiry
ISO 13485	Held or not held. **Not required and not wanted** – see QP-EEG-010 section 1.2
IPC-A-610 class 2 operator certification	How many certified operators, certificate references. IPC-6012 class 2 and IPC-A-600 class 2 apply to the bare four-layer board and IPC-A-610 class 2 to the assembly; all three are required
Multilayer capability	Four-layer FR-4, 1.60 mm ± 10 %, through vias only at 0.60 mm pad on a 0.30 mm finished hole. No blind, buried, back-drilled, filled or plugged vias are used or accepted

Question	Answer required
ESD control programme	ANSI/ESD S20.20 or IEC 61340-5-1, with the EPA boundary described. This matters directly for the ADS1299 front end, the sixteen input networks – 68 kΩ since ECO-EEG-024 was applied on 2026-09-02, 47 kΩ before it – and U1–U3
Any other audited standard	Named

3. The lithium cell

This section governs every kit despatch. It is new in package_v2.2; the v1 package never mentioned lithium transport once.

Scope of this section, stated once. REG-EEG-012 section 3 states **the obligation**: what the article is, how it classifies, what evidence must exist, what conditions the consignment must satisfy and who must be trained. **The procedure lives in PKG-EEG-015 section 7** – the packing steps, the tick sheet, the artwork, the carrier terms, the return-label mechanics and the depot storage rules – and is not restated here. Production evidence that the obligation was actually discharged on the unit that left the building is TST-EEG-004 Rev C step **T29**. Where this section and PKG-EEG-015 section 7 could drift, this section states the rule and PKG-EEG-015 states how it is carried out.

3.1 The article, classified

Property	Value
Cell	Protected 18650 Li-ion, one per kit, kit BOM Rev B item 11
Reference parts	Panasonic NCR18650B in a protection sleeve, or Samsung INR18650-35E in a protection holder
Nominal voltage	3.6 V
Rated capacity	≥ 3000 mAh required by RFQ E-22; 3400 mAh (NCR18650B) or 3500 mAh (35E) as quoted
Watt-hour rating, *calculated*	3.6 V × 3.4 Ah = **12.24 Wh**; 3.6 V × 3.5 Ah = **12.6 Wh**
Article	A single **cell**, not a battery. Under 20 Wh, so section II of the packing instruction applies
Cells per kit	1, installed at J13 in the **HM-10 keyed cell carrier**, behind the **HM-08 quarter-turn battery hatch**. HM-07 is the boom microphone arm and has nothing to do with the cell (PARTS-EEG-019 section 3.1)
Spare cells per kit	**None.** See the decision box below
Proper shipping name	Lithium ion batteries contained in equipment

Property	Value
UN number	**UN3481**
Class	9 (miscellaneous), lithium battery

Decision, recorded. DSN-EEG-002 Rev E section 7 said “a charged spare travels in the case, so a flat cell never ends a session”. That spare has been **removed**. A loose charged cell in the foam changes the consignment from UN3481 *contained in* equipment (PI967) to UN3481 *packed with* equipment (PI966), with different packing, marking and quantity rules, and it puts an unprotected Li-ion cell in a stranger’s home. The compensating controls are RFQ E-22’s four hours of recording at 1000 Hz against a session of about two hours, and the 5 V 2 A charger of A-07 in the case. **RFQ-EEG-001 Rev E S-09 matches this box**: it ships every kit as UN3481, cell inside equipment, and names no spare. **DSN-EEG-002 Rev E does not.** Its section 7 still carries the sentence quoted above, and ECO-EEG-016 section 1.1 records that the file was not reissued in this correction round, so it is wrong on the spare exactly as it is wrong on the boom preamplifier. Where they disagree this box governs, RISK-EEG-011 section 6.1 rules the same way, and DSN-EEG-002 section 7 is corrected at its next issue. The foam bay legended `SPARE CELL` stays in the CASE-00 insert – Rev C keeps it – and travels **empty** in circulation; PKG-EEG-015 sections 2.2 and 7 govern that bay and its tag. Anyone reinstating the spare must re-open this section and PKG-EEG-015 section 7.

3.2 Cell qualification evidence

Held before the first purchase order is released, and re-checked whenever the cell part number changes. This closes RFQ S-04 and the audit finding `cell-qualification-evidence`.

Evidence	Standard or source	Held by	Acceptance
UN 38.3 test summary	UN Manual of Tests and Criteria, sub-section 38.3, summary format of 38.3.5 – mandatory to make available since 1 January 2020	Programme, copy to manufacturer	Must name the cell manufacturer and contact, the test laboratory and contact, a unique test report identifier, the date of the summary, a description of the cell (mass, Wh, chemistry), the list of tests **T.1 altitude, T.2 thermal, T.3 vibration, T.4 shock, T.5 external short circuit, T.6 impact or crush, T.7 overcharge, T.8 forced discharge** with results, the revision of the Manual it was tested to, and a signature. A summary missing any of these is rejected
Safety certificate	IEC 62133-2:2017 + AMD1:2021 certificate or full test report	Programme	Certificate number, body, scope naming the exact cell model. UL 1642 accepted in addition, not instead
Safety data sheet	Cell manufacturer SDS	Programme, carried with each consignment	Current version, in English, with the emergency telephone number
Protection circuit specification	Cell or holder supplier datasheet	Programme	Over-charge cut-off, over-discharge cut-off, over-current and short-circuit trip and recovery, all as numbers. The PCM is a protection means, not the only one: the bq24074-class charger and the two independent interlocks of RFQ S-01 are the others
RoHS and REACH declarations	Section 2	Programme	Per section 2.2

Evidence	Standard or source	Held by	Acceptance
Certificate of conformity	Per lot	Manufacturer	Lot code, date code, quantity, recorded per unit in the device history record per RFQ section 10
Incoming inspection	QP-EEG-010	Manufacturer	Open-circuit voltage at receipt, wrapper and vent inspection, date code, lot

Open, and it is a hardware hole, not a paperwork hole. Two requirements, at two different statuses. RFQ **S-04** requires thermistor-monitored charging. RFQ **E-23**, restored at Rev D and carried in **RFQ-EEG-001 Rev E**, requires a charger IC with thermal regulation and **no charging above 45 °C**. Cite S-04 for the thermistor and E-23 for the temperature; they are two requirements, not one, and the ruling that separates them is RUL-EEG-021 section B.

S-04 is not met, and it stays not met. There is **no NTC net in `design.py`**. The eight ways of J12 are VBAT, DGND, VBUS_CHG, CHG_CE, SDA, SCL, VSYS and NC_CHG_STAT, and J13 is a two-way cell connector carrying VBAT and DGND, so no thermistor way exists on either connector in the Rev B netlist. Nothing measures the cell's temperature, and no test can be written that would.

E-23 is met only in part, and the missing part is the part that matters here. The bq24074-class charger module at J12 regulates its own die temperature, so the "charger IC with thermal regulation" half of E-23 is met on the module. The 45 °C inhibit rests on that same die-temperature regulation alone: it protects the charger, not the cell, and with no NTC the charger cannot know the cell's temperature. TST-EEG-004 Rev C step T4 records that the 45 °C inhibit is not tested and cannot be. So E-23 is met in part, is not verified, and is not closed.

Closure is either a thermistor added to the charger module's temperature-sense pin by ECO, or the 45 °C limit met another way with the alternative written down. The open item is carried in DSN-EEG-003 section 11 and in RISK-EEG-011. Owner: programme technical lead, before Phase 2.

3.3 The three shipping configurations

The rules below are stated as understood at **IATA DGR 67th edition (2026)** for air, **ADR 2025** special provision 188 for road, and **IMDG amendment 42-24** for sea. Editions change every January. The programme's trained dangerous-goods shipper re-verifies this section against the edition in force before each phase's despatch, and records the check.

	A: manufacturer to Brussels	B: Brussels to participant	C: participant to Brussels
Contents	2 to 50 kits in outer cartons, one cell installed per kit	One kit, one carton, one cell installed	Same kit returned

	A: manufacturer to Brussels	B: Brussels to participant	C: participant to Brussels
Classification	UN3481, PI967 section II	UN3481, PI967 section II	UN3481, PI967 section II
Legal shipper	Manufacturer	Programme	**The participant.** This is the hard case
Packages per consignment	More than two	One	One
Lithium battery mark	**Required** (more than two packages)	Not required by the small-consignment relief; **applied anyway** as programme policy, because couriers' own acceptance conditions vary and an unmarked lithium package gets refused at the counter	Applied at despatch and reused on the return leg
Shipper's declaration for dangerous goods	Not required for section II	Not required	Not required
Mode	Air, sea or road as quoted. Sea is cheapest and slowest; the DG rules are equivalent	Road courier within the EU	Road courier

The "applied anyway" policy is one decision, written here as the rule and in PKG-EEG-015 section 7 as the packing step. Neither document may state it differently: the mark goes on **every** carton on **every** leg, whether or not the relief would allow its omission.

Configuration C is the one that has to be engineered, not just documented. The participant is legally the shipper of a dangerous good and cannot discharge a shipper's obligations: they are not trained, they cannot classify, and several carriers will not accept lithium over a retail counter at all. Options considered:

Option	Verdict
(a) Prepaid, pre-classified courier return label in the M-07 carton pocket, on a programme account with a lithium-approved service; carton already carries the lithium battery mark from the outbound leg; participant seals the carton and hands it over	**Adopted.** The participant writes nothing, classifies nothing and declares nothing. The mechanics are in PKG-EEG-015 section 7
(b) Courier collection booked by the programme from the participant's address	**Adopted as the fallback** where the participant cannot reach a drop-off point, and as the default for any kit reported damaged
(c) Participant removes the cell through the HM-08 battery hatch and keeps it; kit returns as non-lithium	**Rejected.** It leaves a loose Li-ion cell in a stranger's home with no protective case and no disposal route
(d) Postal despatch	**Rejected.** Universal Postal Union and Belgian postal rules restrict lithium in the post, and international post is largely closed to it. The programme uses a courier account, not the post

3.4 Section II conditions the packed kit must meet

Every one of these is a condition of shipping, not an optional good practice. **PKG-EEG-015 section 7 turns the whole of this table into a packing procedure with a tick sheet and a sign-off, and TST-EEG-004 Rev C step T29 confirms on the closed kit that it was done.** Neither the procedure nor the tick sheet is repeated here.

Condition	How this kit meets it
Cell passed UN 38.3	Section 3.2 evidence held; summary made available on request
Cell ≤ 20 Wh	12.24 or 12.6 Wh *calculated*
Cell installed in the equipment	At J13, in the HM-10 keyed cell carrier, behind the closed HM-08 battery hatch
Equipment protected against short circuit	The HM-10 carrier cannot accept the cell reversed; the PCM is on the cell; the charge input at J24 is behind a 1.1 A PTC (F1) and a transient suppressor (D24)
Equipment protected against accidental activation	The instrument starts only on a deliberate button press. There is no auto-start path. SW1–SW3 sit under their caps and silicone boots, inside a closed pod, inside foam, inside a closed IP67 case, inside a sealed carton. There is **no hatch interlock** anywhere in this design and none may be claimed

Condition	How this kit meets it
Secured against movement in the outer packaging	The CASE-00 **Rev C** die-cut PE foam insert of PKG-EEG-015 section 2.2 – seven loose-laid 25 mm layers, 175 mm of stack, nine bays, every bay at least 3 mm larger than its part on every axis – with the kit inside the case and the case inside the double-wall carton per M-07. RFQ M-05 requires a die-cut or laser-cut closed-cell PE insert and does not fix the layer count. The two-sheet insert this row named at the first issue of this revision is superseded: PKG-EEG-015 section 2.4 shows it cannot hold the helmet, the pod or the headphones at any arrangement. The seven **Rev C cut files are drawn and shipped** in `mech/` – `CASE-00_foam_layer_1.dxf` to `_7.dxf`, the two Rev B sheets deleted – and they are **not released for cutting** until the bought shell is measured (PKG-EEG-015 section 3.2), so **no insert has been cut** and this condition is met on the schedule and not yet on an insert
Strong rigid outer packaging	Double-wall carton, M-07. **The Nanuk or Peli case is not the shipping container** and M-05 says so
1.2 m drop test on the completed package	**Not yet performed.** See section 3.6
Lithium battery mark	Section 3.5
Net quantity limit per package	One 48 g cell against a limit measured in kilograms. Not approached; the shipper confirms the exact limit in the edition in force
Trained shipper	Section 3.7

3.5 The marks the regulation requires

The obligation is here; the artwork, its dimensions on the printed sheet, its placement on the carton and the label set it belongs to (ART-LBL-06) are in PKG-EEG-015 sections 4.2 and 7.

Mark	What the regulation requires
Lithium battery mark	Rectangular, hatched border, **120 mm wide × 110 mm high**, reduced to not less than 105 mm × 74 mm only where the package is too small for the full size. Carries the text “UN3481” and a telephone number for additional information
Telephone number in the mark	A number that is answered, by someone who can give information about the shipment, during the carrier’s operating hours. **Not a voicemail box.** The programme operations number is used

Mark	What the regulation requires
Class 9 lithium battery hazard label	**Not applicable** to section II consignments and not applied. Applying it wrongly implies a fully regulated shipment and invites refusal
Case marking	Nothing dangerous-goods related goes on the Nanuk or Peli case. The case circulates; the carton is consumed

Any change to the telephone number is an ECO, because it changes a regulatory mark.

3.6 Package qualification

PI967 section II requires the completed package to withstand a 1.2 m drop in the orientation most likely to result in damage, with no damage to the cell, no shifting of contents that would allow contact between the equipment and the cell or between packages, and no release of contents.

This test **has not been performed**. It is a Phase 1 activity, run once on a fully packed kit in its production carton, and the record (date, drop orientations, mass of the packed carton, photographs, post-drop inspection of the cell and of the kit) is retained for the life of the packaging design. It is distinct from the RFQ M-04 1 m drop test of the instrument itself: M-04 asks whether the kit still works, this asks whether the package is legal to ship. Both are run in Phase 1 on the same unit, in that order, and both are recorded. The carton design the test qualifies is PKG-EEG-015 section 6, and it is sized for the enlarged POD-P1 (163.0 × 143.0 × 58.0 mm external) inside the travel case; a package qualified against the v1 pod is not evidence for this one.

3.7 Documents held and training

Held by the shipper for every consignment: the UN 38.3 test summary for the cell; the cell safety data sheet; the package drop-test record from section 3.6; the packing tick sheet signed by the packer; the consignment note or air waybill; and, where the carrier requests it, a written statement that the consignment is UN3481 in compliance with section II of PI967. Retention: three years, or the carrier’s stated period if longer. PKG-EEG-015 section 7 lists the same set as a per-shipment checklist and is the document the packer works from.

Who is the trained shipper. Under IATA the shipper’s personnel must be trained under a competency-based training and assessment programme, with reassessment at intervals not exceeding 24 months. Under ADR chapter 1.3 the consignor’s personnel must have function- specific awareness training, refreshed periodically.

Role	Person	Training	Refresher
Manufacturer’s shipper, configuration A	Named in the quotation. **RFQ-EEG-001 Rev E asks the bidder to name them**	IATA CBTA or national equivalent	Bidder states

Role	Person	Training	Refresher
Programme shipper, configuration B and the return leg's classification	Named individual at the programme, appointed before the first Phase 2 despatch. **Not yet appointed**	IATA CBTA plus ADR 1.3	24 months
Participant	None, and none is required. Configuration C is engineered so the participant never classifies, marks or declares anything	–	–

3.8 State of charge, storage, damaged cells and end of life

State of charge. RFQ S-09 requires kits to be despatched at **≤ 30 % state of charge**. Stated plainly: PI967 section II does not itself impose a state-of-charge limit; the 30 % limit in the regulations belongs to UN3480 cells and batteries shipped alone by air under PI965. The programme adopts ≤ 30 % as an **additional self-imposed control**, because a cell at low charge in a thermal-runaway event releases less energy and because some carriers apply the limit to all lithium consignments in their own conditions of carriage. The MAX17048 gauge reading is taken and recorded at despatch under the PKG-EEG-015 section 7 procedure, and the participant's first instruction in IFU-EEG-014 is to charge the kit before the first session, with the helmet never worn while the charge cable is connected (RFQ S-01, whose two independent interlocks are specified once in that requirement and are not restated here).

Situation	Rule
Despatch, all configurations	≤ 30 % SoC, gauge reading recorded per serial
Storage between participants	Cell at approximately 3.7 to 3.8 V, kit in the case, room temperature, checked every 3 months

Situation	Rule
Cell replacement	**Annually on age, whatever the condition**, and at any refurbishment where the bench capacity check reads below 80 % of rated, the rested open-circuit voltage is below 3.5 V after a full charge, the wrapper is swollen or deformed, the protection module is damaged, the kit has been through a reported drop with the cell in it, or the cell's own date code is five years old – whichever comes first. That list is **SVC-EEG-013 section 2 R9**, which owns it; this row cites it and no longer states a second policy of its own. The "2 years in the fleet" carried here at the first issue of this revision is withdrawn: SVC-EEG-013 section 3.4 already replaces the cell annually, and one cell cannot have two replacement ages. The 80 % figure is **not a gauge reading** – the MAX17048 is a voltage-based ModelGauge part and reports state of charge, not capacity – so it is measured off the unit on a bench cell analyser at the every-fifth-turnaround check of SVC-EEG-013 section 3.2, and it is the only trigger in the list that needs an instrument the turnaround bench does not use every time
Swollen, dented, leaking, deeply discharged or heat-damaged cell	**The kit is never posted or couriered.** Damaged or defective lithium cells are forbidden for transport by air and are subject to special provisions by road. The participant is instructed to stop, not to charge it, and to telephone the programme; the programme arranges a compliant collection or disposal locally
Disposal	Belgian battery collection scheme (Bebat) under Regulation (EU) 2023/1542. **A participant never disposes of a cell.** End-of-programme fleet disposal is a single documented consignment

4. EMC

4.1 Scope, honestly stated

RFQ S-06 asks for a **CISPR 11 group 1 class B pre-scan on one Phase 1 unit**. This is a pre-scan, not a certification: no notified body, no declaration, no CE mark. IEC 60601-1-2 is the EMC collateral standard a medical device would be tested to, and the programme is deliberately **not** claiming it. Immunity is not tested at all in Phase 1; section 4.5 records why and what the decision costs.

4.2 The equipment under test, defined exactly

One Phase 1 unit, complete: routed **four-layer, 150.0 × 130.0 mm** EEG-CAR-01 Rev B carrier with **all thirteen module assemblies** of the twelve purchased types fitted – twelve of them, including both ADS1299 breakouts, on the MP-01 plate (146.0 × 126.0 × 3.0 mm) on M3 × 18 mm nylon standoffs, and the ESP32-S3-DevKitC-1 inserted directly into J6/J7 as the thirteenth; POD-P1 closed (163.0 × 143.0 × 58.0 mm external); the 12-

way screened electrode cable at J14 and the 10-way light cable at J30 both connected to a printed HM-01 with electrodes fitted to a saline head phantom; boom microphone on its pigtail at J18; headphones in the 3.5 mm jack; microSD card fitted at J20; the host USB socket on the ADuM4160 module, presented through its gasketed aperture in POD-P1, connected through a 2 m cable to a mains-powered laptop; battery installed.

Two notes on the EUT that change the result if they change:

- The board is four-layer with continuous AGND_REF and DGND reference planes on L2 and L3. A pre-scan on a two-layer board is not evidence for this design, and if the layer count ever changes back the pre-scan is repeated.
- **No EUT exists yet.** The Rev B routing meets all three conditions of the ECO-EEG-016 section 3 gate – zero design-rule violations, every net one connected copper island, both inner planes continuous under the analogue zone – so the fabrication data is released for review; it is not released for fabrication until the layout review of RFQ-EEG-002A is done (section 9, item 10), and no carrier has been made. If that review moves the plane geometry under the analogue zone or the routing of the host-side nets, the pre-scan is run on the routing as finally released and the report names the Gerber revision it was run on.
- **The host connector is not settled.** RFQ E-24 asks for USB-C, and the named candidate isolator module presents USB-B. The interim answer is the short USB-B-to-USB-C panel pigtail WH-09, and this is a **live non-conformance**, not a settled design. The pre-scan is run on whichever arrangement is fitted, and the report states which, because a panel pigtail is a radiating structure.

Configuration variables fixed and recorded before the first sweep, because changing any of them changes the result: harness routing and dress, cable lengths, laptop on mains versus battery, table height, and the three orientations measured.

4.3 Operating modes measured

Each mode loads a different set of clocks, and a pre-scan that measures only one of them proves nothing.

Mode	What is running
M1 Idle	Powered, no session. ESP32-S3 at 240 MHz, ADS1299 clocks running
M2 Recording to host and card	1000 Hz, 16 channels, USB bulk out plus one-bit SDMMC writes. The frame payload is **50.7 kB/s** (1015 bytes every 20 ms); RFQ E-20's ≈70 kB/s is the allowance once STATUS and SIGNATURE frames and filesystem overhead are added, and it is the figure the card must sustain. The worst case
M3 Audio stimulus	Recording plus I2S to the ES8388 and the headphone amplifier driving the shipped headphone load. RFQ A-04 is restated as 32 to 64 Ω and the shipped ATH-M20x is **47 Ω**, so the bench load is 47.0 Ω
M4 Lead-off excitation	The ADS1299 lead-off current sources active during set-up

Mode	What is running
M5 Contact lights	Shift register clocking, eight bicolour LEDs alternating at **240 Hz** (`LIGHT_PHASE_HZ`) for amber. **Corrected 2026-09-02: the bicolour phase driver is written**, so M5 can be run as specified once a unit exists. Until that date `lights_write()` and `lights_task()` were on/off only and this row said the mode could not be run. The driver alternates a green and a red phase from both halves of the converter's lead-off word; the half-phase quantises to the FreeRTOS tick, so the emitted alternation is **about 250 Hz rather than exactly 240**, which is the figure the EMC engineer should expect to see in the plots. No unit exists, so **M5 has never been run**
M6 Charging, no session	VBUS present at J24, buck-boost and charger switching, session refused by both interlocks

4.4 Ranges, detectors and the pass criterion

Measurement	Range	Detector	Limit
Conducted, on the host USB lead and on the charge lead in M6	150 kHz to 30 MHz	Quasi-peak and average	CISPR 11 group 1 class B
Radiated	30 MHz to 1 GHz	Quasi-peak	CISPR 11 group 1 class B
Radiated, extended	1 to 6 GHz	Peak and average	Only if the radio is ever enabled, which it is not. Run once in M2 anyway as evidence for section 4.6

Pass criterion: worst-case margin of at least **6 dB** below the class B limit in every mode. A margin is required rather than a bare pass because the pre-scan is on one hand-built unit and the fleet will differ. A result between 0 and 6 dB of margin is a finding, not a failure, and goes to the safety reviewer with a remediation proposal. The report states a number, not a picture.

If it fails, the fallback order is: harness dressing and ferrite on the electrode cable at J14; a ferrite on the host USB lead; then a shielded or conductively coated POD-P1. Who pays for the re-scan is settled in the purchase order, not here.

4.5 Immunity

Not tested. The decision is recorded rather than left silent, because a device that must not drop frames during a two-hour session in an arbitrary home is an immunity problem. Items considered and their disposition:

Item	Disposition
ESD to the J15–J17 DIN 42802 sockets and to the exposed host USB connector	Not tested. Mitigated by design: touch-proof sockets, the D1–D16 **BAV99** clamps (BAT54S is not approved in these positions, because Schottky leakage across the series resistor is an offset error on a 10 µV input), the D23 clamp on the comparator output, and the ADuM4160 barrier. Recorded as a residual risk in RISK-EEG-011
Radiated RF immunity, home Wi-Fi and mobile handsets	Not tested. The signal band is DC to 100 Hz and every electrode lead carries the series-resistor and 10 nF low-pass whose corner, flatness and Johnson-noise arithmetic are computed once in RISK-EEG-011 section 4 and are not recomputed here. That network is the principal defence
Mains transients coupled through the host	Not applicable during recording: the isolator is the only path and the instrument is battery powered (RFQ S-01, S-03)
Effect of any immunity event on the data	The F-06 ring buffer and the F-07 GAP frame mechanism mean a disturbance shows up as a declared gap rather than as silently corrupted data. F-06 is relaxed to **90 seconds** of ring plus unlimited backfill from the microSD copy (ECO-EEG-025), because the mandated module's 8 MB of PSRAM cannot hold the three minutes the requirement originally asked for. That is the honest mitigation and it belongs in the analysis plan, not only here

4.6 Proving the radio is never initialised

RFQ S-06 says production must confirm the radio is disabled. The v1 package's only evidence was a comment in `main.c`. The evidence chain is now three links, and all three are required:

1. **Build.** `sdkconfig.defaults` excludes the Wi-Fi and Bluetooth components from the image. No call to `esp_wifi_init`, `esp_bt_controller_init` or `nimble_port_init` exists in the source, and the release build fails if one appears. The firmware release record carries the image SHA-256.
2. **Image identity.** The hash flashed at provisioning (RFQ F-18) is recorded per serial in the calibration record, so the image on any unit in the field can be tied back to the build that was scanned.
3. **Production test.** A 2.4 GHz receiver or spectrum sweep, 60 s during a recording block, accepting only if no carrier above the receiver noise floor, no ESP-prefixed SSID and no BLE advertisement is seen from the unit under test. This is the step S-06 demands and TST-EEG-004 Rev B did not contain; it is **T24, "Radio silent"**, in TST-EEG-004 Rev C, which owns the step numbering, and it runs concurrently with T14 so it costs no line time. It records pass or fail, the receiver noise floor and the firmware image hash.

5. IEC 60601-1 as a design reference

IEC 60601-1:2005 + A1:2012 + A2:2020 is used as a **design reference** for a type BF applied part. The applied parts are the eight scalp cup electrodes, the two ear-clip references, the bias electrode and the three EMG snap electrodes: **fourteen patient terminations** (8 + 2 + 1 + 3), all connected through the 12-way screened electrode cable at J14 – terminations J14.1 to J14.11 – or the three DIN 42802 sockets at J15–J17. The two EOG spare channels are protected on the carrier and brought to J22, but they are **not fitted to panel sockets in a standard build**, so they are not applied parts of the standard build and are not counted here.

The table is a gap list, not a conformity claim. "Met by design" means the design does the thing the clause asks for; it does not mean anyone has measured it. Step numbers are cited from TST-EEG-004 Rev C, which owns them; none is invented here.

Clause	What it asks	Status	Evidence, or what is missing
4.3 Essential performance	Identify it	**Partly.** Essential performance is named in RISK-EEG-011 as: acquire and record the sample stream without undeclared loss, and not exceed the S-02 auxiliary current	Named, never verified against a fault set
4.7 Single fault condition	Safe in single fault	**Met on the calculated budget, corrected 2026-09-02.** The architecture is unchanged – BAV99 clamps to AVDD/AVSS, battery-only recording, the isolator – and the series resistance is now **68 kΩ per lead**, which puts the single-fault DC figure at **36.8 µA against S-02's 50 µA limit**. Until that date this row read "not met on the calculated budget … 53.2 µA against S-02's 50 µA limit, a calculated failure, not a pass", with 47 kΩ fitted	ECO-EEG-024 is **applied** in `design.py`; RISK-EEG-011 section 4 owns the arithmetic and **has not been re-issued for 68 kΩ, so it still prints 53.2 µA as the live state**. A met calculation is not a safe device: **no fault-insertion testing has been done**, nothing has been measured, and the electrical safety reviewer has not seen it

Clause	What it asks	Status	Evidence, or what is missing
7.2, 7.4 Marking and identification	Legible, durable marking	**Met by design.** ART-LBL-01, 50 × 25 mm matt polyester, IPA-resistant, per ASM-EEG-007 stage 6	No IEC 60601-1 symbol set is applied. A certified device would carry the type BF symbol; this one deliberately does not, because carrying it would imply conformity
8.4.2 Patient auxiliary current	≤ 10 µA DC, ≤ 100 µA AC normal; ≤ 50 µA DC single fault	**Calculated only, and the single-fault limit is now calculated as met** – 36.8 µA against 50 µA at the 68 kΩ of ECO-EEG-024, which is applied; see clause 4.7. Corrected 2026-09-02: this cell read “calculated as exceeded (53.2 µA against 50 µA)”. RFQ S-02, budget in RISK-EEG-011 section 4, which still prints the 47 kΩ figure	Never measured. **TST-EEG-004 Rev C step T23** is the routine per-unit measurement, at all **fourteen** terminations – J14.1 to J14.11 and J15, J16, J17 – recording 14 DC and 14 AC values normal, host connected and disconnected, plus 14 single-fault DC values. T23 is explicitly a **stand-in**: a 100 kΩ measuring resistor, not the IEC figure-12 measuring device, and one single-fault condition only
8.5.1 Means of protection	Two means of patient protection to mains	**Met by design.** During recording the instrument is battery-only and the only path to a mains-referenced host is the ADuM4160 module, rated ≥ 2.5 kV RMS for one minute	RFQ S-03, E-24. The 2.5 kV RMS figure is the **module supplier’s type-test certificate**, collected once at incoming inspection (TST-EEG-004 Rev C step T00) and never repeated per unit

Clause	What it asks	Status	Evidence, or what is missing
8.5.2, 8.9 Creepage and clearance	8 mm creepage / 5 mm clearance for 2 MOPP at 250 V working	**Met by design, and the keep-out is evidenced rather than asserted.** The barrier is entirely inside the module. The carrier has no copper on the host side: the isolation keep-out defined in DSN-EEG-003 section 3.3 – the strip x ≥ 141 mm, y = 2 to 22 mm, which is 9 mm wide at the board edge – has to be free of copper on **all four layers**, L1, L2, L3 and L4, not merely on the two outer ones. Both inner layers carry reference planes, so the keep-out has to be cut in the planes as well, and that is the change the four-layer decision forces on this clause. It is the one thing in this row that is a fact about the artwork rather than an intention, so it is not asserted here; it is read off the DRC report	**The DRC report is the evidence.** `kicad/EEG-CAR-01_RevB_DRC_report.txt` tests the strip separately on F.Cu, In1.Cu, In2.Cu and B.Cu and reports **zero isolation keep-out violations**, so the strip is clean on all four copper layers as at the Rev B routing. That is stated here because the report says it. Two qualifications travel with it: that report is an automatic check on a routing no layout engineer has reviewed, so the artwork is released for review and not for fabrication and can still move (section 9, item 10); and nothing has been measured on a fabricated board. The FAI in QP-EEG-010 measures it, and TST-EEG-004 Rev C verifies the keep-out from the artwork once per routed Gerber revision, not per unit
8.5.2 continued	No second path across the barrier	**Residual risk, controlled by assembly.** The ESP32-S3-DevKitC-1's own two USB-C sockets sit on the non-isolated side and bypass the barrier entirely if reached. The DevKit's UART USB-C port is also the end-of-line flashing port, so it must be reachable on the line and unreachable in the field	ASM-EEG-007 stage 6 requires the assembler to confirm they are not reachable from outside a closed pod, and stage 6 records it. This is a **procedural** control on a safety-critical property, which is weaker than a physical one, and the safety reviewer is asked to rule on it

Clause	What it asks	Status	Evidence, or what is missing
8.7 Leakage currents	Measured	**Partly, by a stand-in.** TST-EEG-004 Rev C step T23 gives routine per-unit leakage evidence at all fourteen terminations	It is not the IEC 60601-1 patient-auxiliary-current measurement and does not claim to be. The real verification is owed by the electrical safety review, which has not happened
8.8 Dielectric strength	Type-tested	**Not type-tested by the programme.** The 2.5 kV RMS figure is the module vendor's rating	Per-unit evidence is **TST-EEG-004 Rev C step T20**, a **500 V DC insulation-resistance measurement** across the barrier with a 60 s dwell and a ≥ 1 GΩ limit. **There is no routine per-unit hipot and none is permitted:** a 2.5 kV AC station on an assembled unit stresses a barrier that was already type-tested by its maker and risks damaging it. Any fixture design that specifies a per-unit 2500 V AC test is superseded by T20
8.10 Components and wiring	Components suited to their application	**Partly.** The isolator and both ADS1299 modules are consumer breakouts with no controlled revision and no vendor commitment, and the J15–J17 touch-proof sockets have no confirmed catalogue part at all	AVL-EEG-017 pins the vendors and requires change notification. Today the strongest statement possible is "not substitutable"
8.11 Mains parts	–	**Not applicable.** No mains part. The 5 V charger is a separate CE-marked appliance and is never connected during a session (RFQ S-01)	–

Clause	What it asks	Status	Evidence, or what is missing
9.2, 9.8 Mechanical hazards	No hazardous moving or sharp parts, adequate retention	**Met by design.** Printed frame, no moving parts other than the ratchet and the HM-08 hatch. 1 m drop per RFQ M-04	M-04 drop test not yet run
10.1 Radiation	Ionising and non-ionising	**Not applicable** for radiation. Acoustic output is the real hazard here and now has its own requirement: **RFQ E-29, ≤ 100 dB SPL at any commanded level**, measured on an artificial ear, with the firmware clamping the codec volume register to the value measured at calibration. Calculated full-scale output is about **110 dB SPL**, which is why the requirement exists	**TST-EEG-004 Rev C step T28** measures it as a type test, listed in TST-EEG-004 section 14, **once per lot, not per unit**, and publishes the clamp register value that T17 reads back on every unit. Not yet performed
11.1 Excessive temperatures	Applied-part and touchable-surface temperature limits	**Not tested.** RFQ 9.2's three-hour thermal run is the first measurement, and no limit is written for the TPU pads against the forehead until then. The DevKit's own 3V3 regulator is a second unmeasured thermal item: the carrier draws a *calculated* 288 mA worst case from it, about 0.5 W inside a closed pod, and TST-EEG-004 Rev C step T3 reports the case temperature	Gap: the applied-part limit must be set before the thermal run, not after. If the regulator case exceeds 85 °C, a 3.3 V regulator on the carrier fed from V5V is an ECO against Rev C
11.6 Ingress of liquids	Suited to the environment	**Met by design.** All pod openings gasketed or recessed (RFQ M-02); gel and saline cannot reach the carrier; IP67 travel case (M-05)	No IP test is claimed for the pod itself

Clause	What it asks	Status	Evidence, or what is missing
11.8 Interruption of the supply	Safe on power loss	**Met by design.** Loss of power ends the session; the microSD holds the authoritative copy and the ring buffer backfills on re-enumeration	The ring is 90 s, not three minutes (F-06 as relaxed by ECO-EEG-025), with unlimited backfill from the card. TST-EEG-004 Rev C step T15 is the backfill test
12.2 Usability	IEC 62366-1 process	**Not done.** No usability engineering file exists. The device is used unsupervised, at home, by a distressed cohort, which is precisely the case the standard exists for	IFU-EEG-014 is written in its place. This is a stated shortfall
13.1 Hazardous situations	Enumerated and controlled	**Partly.** RISK-EEG-011 is a risk analysis. It is not a full ISO 14971 risk management file with a plan, a management review and a production-and-post-production feedback loop	Named as a shortfall in RISK-EEG-011 itself

Clause	What it asks	Status	Evidence, or what is missing
14 PEMS	IEC 62304 software lifecycle	**Not done.** The firmware is developed under version control with a released image hash, and that is all. A lifecycle is a process, and building an image is not one	Corrected 2026-09-02. The firmware **is built** – ESP-IDF v5.2.5, esp32s3, four images and their SHA-256 in `firmware/release/manifest.json` – and has **booted once under QEMU**, which emulates none of this design's peripherals. **It has never run on hardware.** The contact-light phase driver **is written**, so T11 is no longer blocked by missing code; it has not been run, because no unit exists. Until that date this cell read "firmware has never been compiled or run on hardware" and "the phase driver is specified and not coded, so T11 cannot pass today"
15.3 Mechanical strength	Drop, impact	**Not tested.** RFQ M-04, Phase 1	–
16 ME systems	Host computer forms a system	**Handled by isolation.** The participant's laptop is a mains-referenced non-medical device separated by the ADuM4160	RFQ F-17
17 / IEC 60601-1-2	EMC collateral	**Not claimed.** CISPR 11 class B pre-scan only, section 4	–

Clause	What it asks	Status	Evidence, or what is missing
IEC 60601-1-11	Home healthcare environment	**Not claimed, and it is the collateral that fits best.** Its requirements on mechanical robustness, environmental range, mains interruption and lay-user instructions are exactly this device's operating case	Recorded as the most significant unaddressed collateral
IEC 60601-2-26, EEG particular	Input impedance, noise, electrode-lead safety, defibrillation	**Partly by design.** Touch-proof 1.5 mm DIN 42802 connectors satisfy the requirement that an electrode lead cannot be inserted into a mains socket. Noise floor *calculated* at **0.31 µV RMS with the 68 kΩ resistors of ECO-EEG-024, which are what is fitted since 2026-09-02** (it was 0.27 µV at the 47 kΩ this row used to name as fitted), against RFQ E-03's 1.0 µV limit; the arithmetic is in RISK-EEG-011 section 4. **Defibrillation protection is not provided and is not claimed**	Never measured. TST-EEG-004 Rev C steps T7 (gain), T8 (noise floor) and T22 (frequency response)

What would have to change if the device were ever certified. All of the following, at minimum: a full ISO 14971 risk management file with production and post-production feedback; an IEC 62366-1 usability engineering file with formative and summative evaluation; an IEC 62304 software lifecycle for the firmware and for the browser session runner; type testing for dielectric strength, leakage, temperature and mechanical strength by an accredited laboratory; IEC 60601-1-2 immunity and emissions testing rather than a pre-scan; controlled vendors with change notification for every safety-relevant component, which the consumer breakouts at J1–J4 and J10 cannot supply today; a sourced and first-articled touch-proof socket at J15–J17; a physical rather than procedural control on the DevKit's own USB sockets; the type BF symbol and the full IEC 60601-1 marking set; a quality system to ISO 13485; and a manufacturer legal entity that takes on the obligations. The programme has no intention of doing any of it, and says so here so that nobody later mistakes the design references for a certification path.

6. Biocompatibility

6.1 Categorisation

Under ISO 10993-1:2018 the instrument is a **surface-contacting device, intact skin**. Per participant, contact is two or three sessions of about two hours, which is *limited* contact (≤ 24 h). The programme adopts the more conservative **prolonged** category (> 24 h to 30 d) because participants may run additional sessions, because the same physical parts pass through a sequence of participants – about twenty a year on one kit, at the twenty turnarounds per kit per year SVC-EEG-013 section 3.5 costs, and more over the life of a fleet built at 25 to 50 kits (RISK-EEG-011 section 1.4) – and because the abrasive scalp preparation the protocol uses means the skin is not always intact by the end of the session. Required endpoints:

Endpoint	Standard
Cytotoxicity	ISO 10993-5:2009
Skin sensitisation	ISO 10993-10:2021
Irritation	**ISO 10993-23:2021.** Irritation testing moved out of ISO 10993-10 in the 2021 revision. A declaration citing “ISO 10993-10 irritation” against the 2021 edition is wrong on its face

RFQ S-05 names three items. The actual skin-contact set is larger, and section 6.2 is that list.

6.2 Skin-contact material list

Eight rows. RFQ A-03’s separate headband or cap is **withdrawn as a kit item**: the eight electrodes are fixed to the HM-01 frame at manufacture, so a headband carrying fixed holders at the same eight sites would duplicate them. A-03 is rewritten to cover the parts the kit actually needs and already has on the frame – the **HM-06 chin strap** and the **HM-03 occipital yoke** – and both are covered by the rows below. RISK-EEG-011 section 2 (Chemical) and its section 7.3 item 4.6 count the same eight materials.

Part	Material or product	Reference supplier	Contact site and duration	Evidence route
Sintered Ag/AgCl cup electrodes, 8 fitted + 2 spare (item 25)	Sintered Ag/AgCl in an HM-04 body	Wuhan Greentek; FRI or Spes Medica	Scalp, 2 h, through paste	Supplier declaration, -5 and -10/-23
Ag/AgCl ear-clip references ×2 (item 26)	Ag/AgCl and clip body	Greentek	Earlobe, 2 h	Supplier declaration
EMG snap leads (item 27) and disposable snap electrodes ×30 (item 28)	Hydrogel snap electrodes	Ambu BlueSensor N or Kendall	Cheek, submental, laryngeal, 2 h, single use	Existing manufacturer declarations. The easiest row in the table

Part	Material or product	Reference supplier	Contact site and duration	Evidence route
HM-02 TPU comfort pads ×4 and HM-06 chin strap and liner (item 19)	TPU 85A, printed or cast	To be named	**Forehead, occiput, crown, chin. The parts most continuously against skin.** Consumable, replaced each turnaround	Material supplier declaration plus the print bureau's process declaration
HM-01 halo and arches, HM-03 occipital yoke, HM-04 bodies (items 15, 16)	MJF PA12, dyed black	Print bureau, to be named	Scalp, forehead and occiput through the pads, and directly where the pads do not cover	**The hardest row.** See section 6.4
HM-07 boom arm and its cheek sleeve, HM-08 battery hatch (items 17, 22)	MJF PA12, gooseneck sleeve	Print bureau; gooseneck supplier	Boom sleeve at the cheek, incidental. The HM-08 hatch is handled rather than worn, and is included because it is the same printed, dyed material	Declaration
Conductive EEG paste ×2 100 g, abrasive prep gel 100 g, saline wipes ×30 (item 31)	Greentek GT20/GT5 or Ten20/NuPrep	Greentek or Weaver	**Applied by the participant, and not all to the same site.** Conductive paste into the eight scalp ports and onto the two ear clips, 2 h; abrasive prep gel on the face, the neck and the earlobes only and never on the scalp; saline wipes at those same three EMG sites. See IFU-EEG-014 section 13.2	SDS plus the manufacturer's existing skin-contact evidence. See section 6.5

Part	Material or product	Reference supplier	Contact site and duration	Evidence route
ATH-M20x ear cushions (item 29)	Protein leather over foam	Audio-Technica	Circumaural, 2 h, both ears. **Never mentioned in S-05**	Supplier declaration or documented history of safe use

6.3 What a valid declaration looks like

The programme sends one page per supplier, and accepts a returned declaration only if it carries all of the following:

1. The supplier's legal entity, address and the signatory's name and position.
2. The **exact material grade or product part number**, matching the AVL-EEG-017 entry.
3. The contact category assumed: surface device, intact skin, prolonged contact.
4. The endpoints addressed, each named with its **standard number and edition year**.
5. The basis: a test report (with laboratory name, report number and date), or a documented history of safe use in the same contact category with the reference given, or a toxicological risk assessment by a named assessor.
6. Any conditions or limits on the declaration – post-processing, colourants, sterilisation method, cleaning agents.
7. A statement that the declaration covers the material **as supplied to this programme**, not a different grade from the same family.
8. Date, and an expiry or recheck date.
9. A named contact for questions.

Reject a declaration that: says only "biocompatible" or "medical grade" with no standard cited; cites ISO 10993-1 alone, which is a categorisation standard and contains no test; cites "ISO 10993-10 irritation" against the 2021 edition (see section 6.1); covers a different grade, colour or supplier from the one on the AVL; is a marketing page rather than a signed document; is undated; carries no signatory; or is silent about colourant and post-processing for a printed part.

6.4 MJF PA12, treated separately

MJF PA12 is not biocompatible by default. The variables that decide it are the black dye, the post-process chemistry and the residual unfused powder, none of which the base polymer's datasheet addresses. The bureau must supply, in addition to the section 6.3 declaration:

- The polymer grade and its supplier, and the fusing and detailing agent chemistry.
- The dyeing process and the colourant, with its own declaration.
- The depowdering method, and a statement on **residual free powder in the HM-01 internal wiring channels**, which is the part nobody looks at and which SVC-EEG-013 flushes with warm water at every refurbishment.
- Any sealing or coating applied, and whether it changes the answer.
- Confirmation that the declaration still holds after **25 cycles of the SVC-EEG-013 reprocessing protocol**, including 70 % IPA wiping. A declaration that is true only as-printed is not sufficient for a part that circulates for years.

If the bureau cannot supply this, the alternative is programme-commissioned ISO 10993-5 and -23 testing on printed, dyed, depowdered coupons from the same build, which is a cost and a lead time the Phase 2 schedule has to carry.

6.5 Consumables applied by the participant

The participant applies the paste, the abrasive prep gel and the saline wipes to themselves, without supervision, and the two gels do not go to the same place: the conductive paste is introduced into the eight scalp ports and onto the two ear clips, while the abrasive prep gel is used on the face, the neck and the earlobes only and never on the scalp, which is what IFU-EEG-014 section 13.2 prints for the participant. The programme holds the SDS for each, retains the manufacturer's skin-contact evidence, and puts the following in IFU-EEG-014 and the participant information sheet:

- Do not use the abrasive gel on broken skin, a skin condition or a recent wound at the face, neck and earlobe prep sites, and do not use it on the scalp at all. Do not use the conductive paste on broken skin or an active scalp condition.
- Stop and remove the headset if the skin stings, burns or itches.
- Contraindication list: known allergy to any listed constituent, active scalp condition, broken skin at any electrode site.
- A support contact that is answered, and that is **not** the research team (ETH-EEG-001).

6.6 Evidence register

One row per material in section 6.2 – eight rows: part, material grade, supplier, declaration reference, revision, date received, endpoints covered, standard editions cited, recheck date, and the reviewer who accepted it. The register feeds hazard H-19 in RISK-EEG-011 and is checked before each phase's purchase order. **The register is empty today.** Not one declaration in section 6.2 has been requested.

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7. Ethics and data protection

7.1 Link to the study

The instrument may not be issued to any participant before an independent ethics committee has approved the protocol. ETH-EEG-001 is the device annex to that submission, written in committee language, and it carries the **configuration freeze record**: the exact revision of every controlled document, the routed Gerber revision, the firmware image hash, the BOM revision and the serial range of units covered by the approval. That freeze record is produced by ECO-EEG-016 from the as-built record.

The study is pre-registered. The pre-registration identifier, the registry and the date go in ETH-EEG-001 section 1 and are quoted on the participant information sheet, so a participant can read the analysis plan that existed before their data were collected. **The identifier is not yet allocated** and is a placeholder until the registration is filed.

Amendment rule. Any Class 1 change under ECO-EEG-016 – a risk control component, an applied-part material, the isolation architecture, the acoustic ceiling, or the cell – is a protocol amendment and requires re-notification to the committee before any affected unit is issued. The ECO form carries the question "does this require re-notifying the ethics committee?" precisely so that this cannot be forgotten. The four-layer carrier and the enlarged POD-P1 are recorded as ECOs in the ECO-EEG-016 register;

they change the isolation keep-out's realisation on the two new inner layers, so they are re-notifiable if a freeze record has already been submitted.

7.2 GDPR position of the recorded data

Question	Position
Controller	The TI One Voice research programme, Brussels
Data	Sixteen channels of EEG, the participant's recorded voice from the boom microphone, room audio from the room microphone during scripted windows only, button responses, and session metadata
Category	**Special category data** under Article 9(1): data concerning health, and biometric-adjacent voice data
Legal basis	Article 6(1)(a) consent, together with Article 9(2)(j) scientific research subject to Article 89(1) safeguards and the Belgian Law of 30 July 2018
DPIA	**Required** under Article 35. Large-scale special-category processing of a vulnerable cohort, with recording equipment operating inside participants' homes. ETH-EEG-001 section 4 is the device part of it
Where the data physically live	The authoritative copy is written to the microSD card at J20 (RFQ E-20) inside a kit that **travels back through the post**. The host copy reaches the programme over TLS during the session
Room microphone	Hardware-mutable on MIC_MUTE (GPIO21) and captured only during scripted windows. This is a control the DPIA relies on, and TST-EEG-004 Rev C step T17 verifies the mute depth
At rest on the card	**Open decision.** The card is not encrypted today. Either encrypt at rest with a key that is not on the device, or record the decision not to and justify it against the loss-in-post scenario. Owner: programme technical lead, before Phase 2
Erasure	Secure erase of the microSD at refurbishment, after ingest is confirmed, recorded on the SVC-EEG-013 refurbishment record. A kit never carries one participant's data into another participant's home
Kit lost in the post	Treated as a personal data breach: assessed within 72 h, notified to the Belgian DPA if the risk threshold is met, and the affected participant told. The mitigation is the encryption decision above
Retention	Per the approved protocol, in ETH-EEG-001

Question	Position
Pseudonymisation boundary	The unit serial identifies an instrument. The format is **`TIOV-B-nnnn`** – programme prefix, hardware revision letter, four digits, with Phase 1 using 0001–0009, Phase 2 0010–0099 and Phase 3 0100–0999 – **defined once in PKG-EEG-015 section 5** and quoted here only so this row reads on its own. It appears identically on the label, in the Data Matrix, in the USB `iSerialNumber`, in the calibration record and on the packing list. The participant identifier is separate, and the mapping between them is held apart from the recordings, by a named person. **Analysis files carry the unit serial, never the participant identifier and never a name**

7.3 The per-device key: what it proves and what it does not

Each unit holds an ATECC608B secure element at J11. Its P-256 private key is generated on-chip and never leaves it. At provisioning (RFQ F-18) the public key is exported and the fingerprint is computed to the single definition in **FW-EEG-001 section 7**, which every other document cites and none restates. The fingerprint is on the label and in the calibration record; the **unit serial `TIOV-B-nnnn`** becomes the USB `iSerialNumber` (F-04, and RUL-EEG-021 section B rules it); the ATECC608B's own nine-byte factory serial is a second identifier, printed and carried in the Data Matrix, and is **not** the descriptor string; the firmware signs each block of the frame stream (F-08).

What a valid signature proves. That the signed block was produced by a unit holding the private key matching a registered public key, and that the block's bytes have not been altered since signing. It ties a stream to a specific physical instrument, and it means a recording cannot be silently edited, spliced or fabricated after the fact by anyone who does not hold that key.

What it does not prove, stated plainly because the temptation to over-claim is real.

- Not who wore the device, or that anyone wore it. The key signs whatever the converters produced.
- Not that the electrodes were on a head. A signed recording of a saline phantom, or of an open input, is exactly as valid a signature as a signed recording of a person.
- Not that the analogue signal is genuine. Anything injected ahead of the ADS1299 is signed along with everything else.
- Not that the recording happened when the timestamps say, unless the timestamp chain is anchored to something outside the device. It is not, today.
- Not that the unit was not opened. There is no tamper detection on POD-P1, and a unit that is lost or stolen keeps signing.
- Not a legal chain of custody, and it must never be described as one in a paper.

The key is an integrity control against post-hoc alteration of study data. It is not an anti-fraud control against a participant, it is not evidence about anyone's experience, and nothing in a publication may imply otherwise. On the Phase 1 prototypes the eFuses are **not burned**: secure boot and flash encryption are enabled from Phase 2, so the two prototypes run unsigned images and the signature chain above is the only integrity control on them.

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8. Compliance matrix

#	Obligation	Applicable document	Responsible	Evidence retained	Status
1	Not a medical device under MDR 2017/745	MDR Art. 2(1); section 1	Programme lead	This section 1, signed and dated; ethics committee determination	**Drafted.** Committee determination outstanding
2	Controlled label and participant wording	RFQ M-03; section 1.4; PKG-EEG-015	Programme lead; manufacturer applies	ART-LBL-01 artwork; IFU-EEG-014	**Wording released.** Artwork in PKG-EEG-015
3	Customs description and HS code	Section 1.5	Programme lead with broker	Commercial invoice template	**Open.** Code unconfirmed
4	RoHS 3 conformity, contractual, covering the four-layer laminate ply by ply	2011/65/EU + 2015/863; EN IEC 63000; RFQ S-08; sections 2.2, 2.3	Manufacturer declares; programme accepts	Signed declaration per phase; supplier declarations; fabricator laminate declaration; exemption list	**Not started.** No declaration received
5	REACH SVHC article-level statement	1907/2006 Art. 33; section 2.3	Manufacturer declares; programme communicates to participant	Declarations with candidate-list version and date; Article 33 statement in IFU-EEG-014	**Not started**
6	SCIP notification decision	ECHA; section 2.4	Programme lead	Recorded decision plus the declarations behind it	**Not started.** Cannot be closed until the J15–J17 socket part is chosen

#	Obligation	Applicable document	Responsible	Evidence retained	Status
7	EU DoC for the 5 V charger and cables	LVD 2014/35/EU, EMC 2014/30/EU; section 2.5	Manufacturer supplies; programme accepts	Vendor DoC, test reports, CE traceable to a legal entity	**Not started.** “Certified generic” is not acceptable
8	WEEE and battery take-back	2012/19/EU; (EU) 2023/1542; section 2.5, 3.8	Programme lead	Registration or disposal consignment records	**Not started.** Due at end of programme
9	Cell UN 38.3 test summary	UN Manual of Tests and Criteria 38.3.5; RFQ S-04; section 3.2	Cell supplier via manufacturer; programme holds	Test summary, all ten required fields, T.1–T.8	**Not started**
10	Cell IEC 62133-2 certificate	IEC 62133-2:2017+A1:2021; section 3.2	Cell supplier; programme holds	Certificate or full report naming the exact model	**Not started**

#	Obligation	Applicable document	Responsible	Evidence retained	Status
11	Thermistor-monitored charging (S-04) and no charging above 45 °C (E-23)	RFQ S-04, E-23; section 3.2 box; RUL-EEG-021 section B	Programme technical lead	ECO adding the NTC net, or a written alternative	**S-04 not met, and it stays not met:** no NTC net exists in the Rev B netlist and no thermistor way exists on J12 or J13. **E-23 met only in part:** the bq24074-class module's own die thermal regulation is the whole of what stands behind the 45 °C inhibit, nothing measures the cell, and TST-EEG-004 Rev C T4 records that the inhibit is not tested and cannot be. Hardware hole, open
12	UN3481 / PI967 section II compliance, configurations A and B	IATA DGR, ADR SP188, IMDG; RFQ S-09; sections 3.3–3.5; PKG-EEG-015 §7	Manufacturer (A); programme (B)	Packing tick sheet per consignment, marks applied, documents per section 3.7	**Obligation released here; procedure in PKG-EEG-015 §7; per-unit evidence is TST-EEG-004 Rev C T29.** Not yet exercised
13	Return leg engineered so the participant never classifies or declares	Section 3.3, option (a)	Programme operations	Prepaid pre-classified label; carrier account terms; collection fallback	**Decided, not yet contracted** with a carrier

#	Obligation	Applicable document	Responsible	Evidence retained	Status
14	1.2 m package drop test	PI967 §II; section 3.6	Programme, Phase 1	Drop record, orientations, photographs, post-drop inspection	**Not performed**
15	Trained dangerous-goods shipper	IATA CBTA; ADR 1.3; section 3.7	Manufacturer names theirs; programme appoints its own	Training certificates, 24-month refresher log	**Not appointed** at the programme
16	State of charge ≤ 30 % at despatch	RFQ S-09; section 3.8	Whoever despatches	Gauge reading per serial on the PKG-EEG-015 §7 tick sheet	**Rule released**
17	Damaged cell never transported	Section 3.8	Participant instructed; programme acts	IFU-EEG-014 wording; collection record	**Wording released**

#	Obligation	Applicable document	Responsible	Evidence retained	Status
18	CISPR 11 class B pre-scan, one Phase 1 unit, on the four-layer carrier in the enlarged POD-P1	RFQ S-06; section 4	Manufacturer arranges lab; programme accepts report	Lab report, EUT photographs, plots per mode M1–M6, worst-case margin	**Not performed.** Plan released here. The EUT cannot be built until the layout review of RFQ-EEG-002A is done and the artwork is released for fabrication. Corrected 2026-09-02: this cell also said M5 could not be run until the contact-light phase driver existed. **The driver exists**, and M5 will emit its alternation at about 250 Hz rather than the specified 240 Hz, because the half-phase quantises to the FreeRTOS tick
19	Radio never initialised, proven in production	RFQ S-06; section 4.6	Programme (build evidence); manufacturer (T24)	sdkconfig, image hash per serial, T24 result per unit	**Not implemented.** T24 “Radio silent” is the step in TST-EEG-004 Rev C
20	IEC 60601-1 design-reference gap list	Section 5	Programme technical lead; safety reviewer signs	This section 5, reviewed and signed	**Drafted.** No safety engineer engaged

#	Obligation	Applicable document	Responsible	Evidence retained	Status
21	Patient auxiliary current measured at the fourteen terminations	RFQ S-02; section 5	Manufacturer, TST-EEG-004 Rev C T23	14 DC and 14 AC values normal, host connected and disconnected, plus 14 single-fault DC values, per serial	**Calculated only, and the calculation now passes: 36.8 µA** single-fault DC against the 50 µA limit, at the 68 kΩ that ECO-EEG-024 fitted on 2026-09-02. Corrected on that date: this cell read “the calculation fails: 53.2 µA … ECO-EEG-024 is the proposed fix”. The ECO is applied; **the measurement is still owed and the safety reviewer has still not seen it**
22	Isolation barrier routine-tested per unit	RFQ S-03; section 5	Manufacturer, TST-EEG-004 Rev C T20	Applied voltage, insulation resistance, dwell, RH, module type and lot, per serial	**Specified, not performed.** 500 V DC insulation resistance, ≥ 1 GΩ. No per-unit hipot is used
23	DevKit USB sockets unreachable in the field	Section 5; ASM-EEG-007 stage 6	Manufacturer	Stage 6 tick	**Procedural control.** Safety reviewer to rule
24	ISO 10993-5 / -10 / -23 declarations, all eight material rows	RFQ S-05; section 6	Suppliers declare; programme accepts	Evidence register, section 6.6	**Not started.** Register empty

#	Obligation	Applicable document	Responsible	Evidence retained	Status
25	MJF PA12 dye, powder and post-process declaration	Section 6.4	Print bureau	Bureau declaration plus 25-cycle statement	**Not started**
26	Consumable SDS and contraindications	Section 6.5	Programme	SDS on file; IFU-EEG-014 wording	**Not started**
27	Maximum acoustic output ≤ 100 dB SPL, with a firmware clamp	RFQ E-29; section 5, clause 10.1	Manufacturer, TST-EEG-004 Rev C T28, once per lot	Two dB SPL values, headphone model and impedance, clamp register value	**Not performed.** Calculated full-scale output is about 110 dB SPL, so the clamp is load-bearing
28	Ethics approval before any unit is issued	Section 7.1; ETH-EEG-001	Programme lead	Approval letter, configuration freeze record	**Not submitted**
29	DPIA	GDPR Art. 35; section 7.2	Programme lead	DPIA document, device section from ETH-EEG-001	**Not started**
30	Encryption-at-rest decision for the microSD	Section 7.2	Programme technical lead	Recorded decision with justification	**Open**
31	Secure erase at refurbishment	Section 7.2; SVC-EEG-013	Programme operations	Refurbishment record per turnaround	**Procedure to be written** in SVC-EEG-013
32	Per-device key claims kept accurate in publications	Section 7.3	Programme lead	This section quoted in the analysis plan	**Released**

9. What is not known

1. **No safety engineer has reviewed this design.** Section 5 is a self-assessment. It blocks use on a person; it does not block quoting. Fabrication is blocked too,

but by the layout review still owed under RFQ-EEG-002A (item 10), not by this section.

2. **No hardware has been built, manufactured, shipped, scanned, tested or declared.** Every row of section 8 marked “not started” or “not performed” is exactly that. This file is the list of obligations, not a record of their discharge. **Corrected 2026-09-02:** this item read “nothing has been built”. The firmware image is now the one exception – it is built and it has booted in an emulator (clause 14) – and it changes nothing here, because no unit exists to run it on and no obligation in section 8 is discharged by it.
3. **The dangerous-goods rules in section 3 are stated as understood at IATA DGR 67th edition, ADR 2025 and IMDG 42-24.** They are re-verified against the edition in force before each despatch by the trained shipper, who is not yet appointed.
4. **RFQ S-04 and RFQ E-23 are two requirements at two different statuses, and neither is closed.** The thermistor required by S-04 does not exist in the Rev B netlist and there is no thermistor way on J12 or J13, so **S-04 is not met and it stays not met**. **E-23 is met only in part**: the bq24074-class charger module regulates its own die temperature, which is the “charger IC with thermal regulation” half of the requirement, but the 45 °C inhibit rests on that alone, nothing measures the cell, and TST-EEG-004 Rev C step T4 records that the inhibit is not tested and cannot be. Until a thermistor is added by ECO, or an alternative is written down, S-04 stays not met and E-23 stays met in part only.
5. **S-02’s single-fault limit is calculated as met – 36.8 µA against 50 µA – and has never been measured. Corrected 2026-09-02**: this item read “calculated as exceeded – 53.2 µA against 50 µA … nothing in this file may be read as a claim that S-02 is met”, with 47 kΩ fitted. ECO-EEG-024 is **applied**: R1-R16 are 68 kΩ on the same footprints, and E-10 moves to the ±1.0 dB branch it already states for this case. What may still not be read into this file is that S-02 has been *demonstrated*: nothing has been measured, the measurement is owed before Phase 2, and **SR-01 is closed in the design and not signed off**. The production simulation SIM-EEG-018 Rev A, re-run on 2 September 2026 after these changes, reports **193 checks passed, 0 failed and 6 known open items**, and the S-02 entry among them is now “SR-01 is closed in the design and not yet signed off”. The other five are the linked image’s static IRAM reporting full with one byte free, the v1 HM-01 mesh being two disconnected bodies, the unreviewed routing, E-27 never having been seen to light, and the two irreconcilable board-current figures. E-11’s low-pass half and E-27’s missing driver, which this item used to list as open, are met.
6. **No maximum acoustic output has ever been measured.** E-29 is new, the calculated full-scale figure is about 110 dB SPL, and the firmware clamp that enforces the 100 dB limit is specified and not yet coded.
7. **No biocompatibility declaration has been requested from any supplier.** The register in section 6.6 has no rows.
8. **The touch-proof sockets at J15–J17 have no confirmed catalogue part.** They are a Class A patient-safety part on every unit, they carry a 12-week first-article lead time, and neither the RoHS nor the biocompatibility question about them can be asked until the part exists.
9. **The pre-registration identifier and the ethics approval reference are placeholders.** They are filled in by ECO before the first unit is issued.
10. **The routing exists and closes; it is released for review, not for fabrication, and it has not been reviewed by a human layout engineer.** It was produced by

the programme’s own constraint-aware autorouter on the four-layer stack-up, and `kicad/EEG-CAR-01_RevB_DRC_report.txt` is the authority for its state. The report records EEG-CAR-01 Rev B routed on four layers, 150.0 x 130.0 mm, with 3 745 track segments and 552 through vias, and one continuous plane island per net on both In1.Cu and In2.Cu. **All 145 nets are fully connected**: none unclosed, none without copper. Every geometric rule passes: the smallest measured clearance is 0.260 mm on F.Cu, 0.275 mm on B.Cu and 0.285 mm on the planes against a 0.20 mm rule; the narrowest conductor is 0.20 mm; the smallest plated hole is 0.30 mm; copper stands 2.00 mm from every non-plated hole; there are no zone crossings; no digital net enters the analogue zone; and there is exactly one AGND_REF-to-DGND bridge and one HARN_SHIELD-to-DGND bridge. The report’s own line is “VIOLATIONS: 0 – none. The board passes every rule listed above”, which is why clause 8.5.2 above is able to cite the report for the isolation strip rather than assert it. That meets all three conditions of the gate in ECO-EEG-016 section 3 – zero violations, every net one connected copper island, both inner planes continuous under the analogue zone – so the data in `kicad/` is **RELEASED FOR REVIEW under RFQ-EEG-002A**. It is **NOT RELEASED FOR FABRICATION** and no board may be ordered from it, because no human layout engineer has read the routing; that review is what fabrication release waits on. Two things belong beside the zero. The router **relaxed 169 connections** to close the board – 36 took a conductor narrower than the 0.25 mm preferred width, 133 kept full width and took a reduced gap, every one at or above the 0.20 mm minimum conductor and gap – and a board that closes at minimum geometry is not the same board as one that closes at preferred geometry, even when every rule passes. And nothing has been fabricated or measured. The creepage argument of section 5, clause 8.5.2 depends on the keep-out being honoured on all four layers: the report says it is, and it is re-verified from the artwork once per routed Gerber revision – including whichever revision the layout review finally releases – before it is relied upon.